

Enterprise Active Directory & IAM Access Control Infrastructure by Samuel Guardado

Overview

This project demonstrates an enterprise-grade Identity and Access Management environment deployed within a Windows Server Active Directory Domain Services ecosystem. The solution establishes Role-Based Access Control following the AGDLP (Account → Global Group → Domain Local Group → Permission) framework, automated user lifecycle provisioning via PowerShell, share-level Least Privilege enforcement, and automated deprovisioning routines.

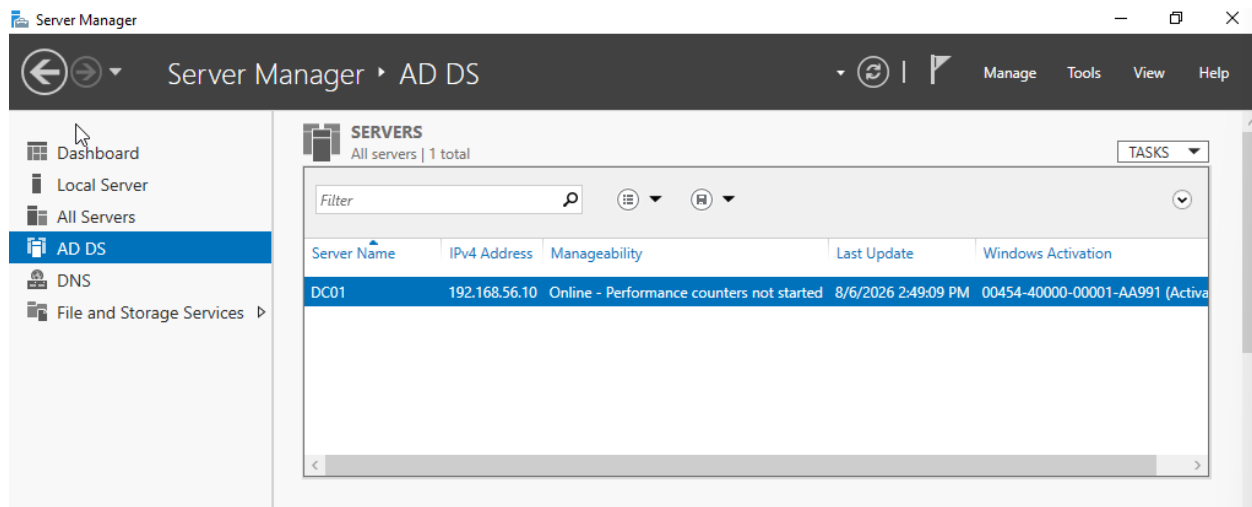
Technical Features & Highlights

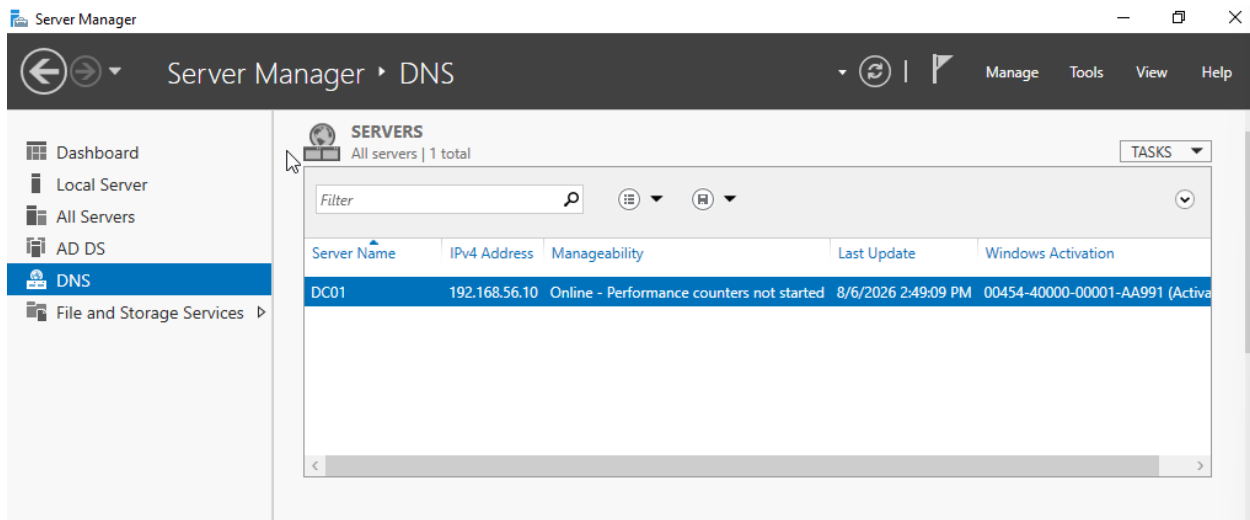
- Identity Architecture: Single-domain Active Directory structure (CORP.LOCAL) running on primary Domain Controller DC01.
- RBAC Standard: Strict AGDLP implementation to decouple individual account identity from direct resource ACLs.
- Automation: PowerShell scripts for bulk account onboarding from HR CSV inputs and instant offboarding/revocation.
- Security Model: Zero Trust Principle of Least Privilege applied to network shares and active accounts.

Implementation

1. Core Domain Infrastructure

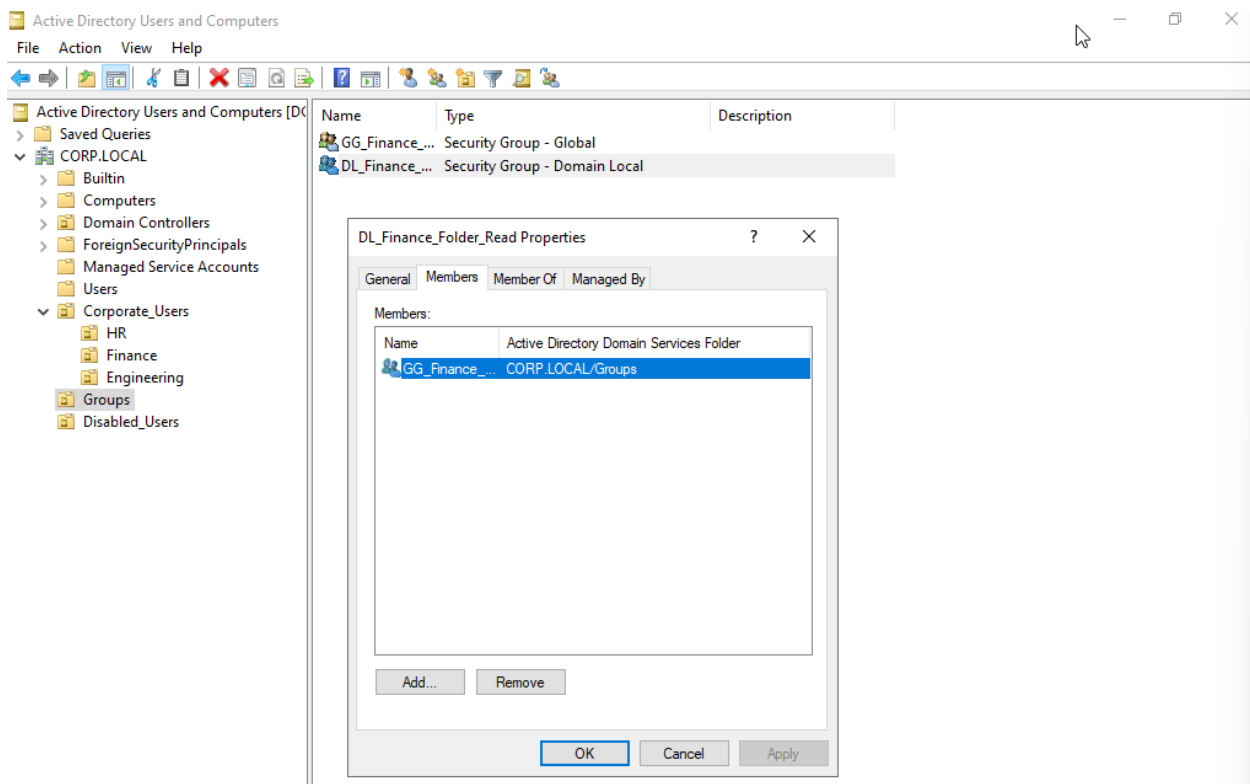
- Installed and validated Active Directory Domain Services (AD DS) and Domain Name System (DNS) on primary Domain Controller DC01 (192.168.56.10).





2. Organizational Unit (OU) Hierarchy & AGDLP Group Structure

- Established an enterprise Organizational Unit structure (Corporate_Users, Groups, Disabled_Users) and configured RBAC adhering to AGDLP standards by nesting Global Group GG_Finance_Read inside Domain Local Group DL_Finance_Folder_Read.



3. Automated Identity Provisioning Engine Execution

- Ingested HR CSV data via PowerShell to dynamically provision Active Directory accounts with standardized sAMAccountName attributes (jdoe, jsmith, arivera), place accounts into designated OUs, enforce mandatory password resets, and assign role-based security groups.

```
PS C:\Users\Administrator>
PS C:\Users\Administrator>
>> New-Item -ItemType Directory -Path "C:\IAM_Scripts" -Force
>>
>> @'
>> FirstName,LastName,Department,Title
>>   >> John,Doe,Finance,Financial Analyst
>>   >> Jane,Smith,HR,HR Specialist
>>   >> Alex,Rivera,Engineering,Systems Engineer
>>   >> '@ | Out-File -FilePath "C:\IAM_Scripts\new_employees.csv" -Encoding ascii
>>
>> # provisioning script
>> @'
>> Import-Module ActiveDirectory
>> $Users = Import-Csv -Path "C:\IAM_Scripts\new_employees.csv"
>>
>> foreach ($User in $Users) {
>>     $SAM = ($User.FirstName.Substring(0,1) + $User.LastName).ToLower()
>>     $OU = "OU= $($User.Department),OU=Corporate_Users,DC=CORP,DC=LOCAL"
>>     $SecurePassword = ConvertTo-SecureString "TempP@ssword2026!" -AsPlainText -Force
>>
>>     if (-not (Get-ADUser -Filter "SamAccountName -eq '$SAM'")) {
>>         New-ADUser -Name "$($User.FirstName) $($User.LastName)" `
>>             -GivenName $User.FirstName `
>>             -Surname $User.LastName `
>>             -SamAccountName $SAM `
>>             -UserPrincipalName "$SAM@CORP.LOCAL" `
>>             -Path $OU `
>>             -AccountPassword $SecurePassword `
>>             -Enabled $true `
>>             -ChangePasswordAtLogon $true `
>>             -Title $User.Title `
>>             -Department $User.Department
>>
>>         Write-Host "Successfully Provisioned: $SAM in $OU" -ForegroundColor Green
>>
>>         if ($User.Department -eq "Finance") {
>>             Add-ADGroupMember -Identity "GG_Finance_Read" -Members $SAM
>>             Write-Host "Assigned $SAM to Security Group GG_Finance_Read" -ForegroundColor Yellow
>>         }
>>     }
>> }
>> '@ | Out-File -FilePath "C:\IAM_Scripts\Provision_Users.ps1" -Encoding ascii
>>
>> # provisioning engine
>> Set-ExecutionPolicy -ExecutionPolicy Unrestricted -Scope Process -Force
>> C:\IAM_Scripts\Provision_Users.ps1
```

```
Directory: C:\

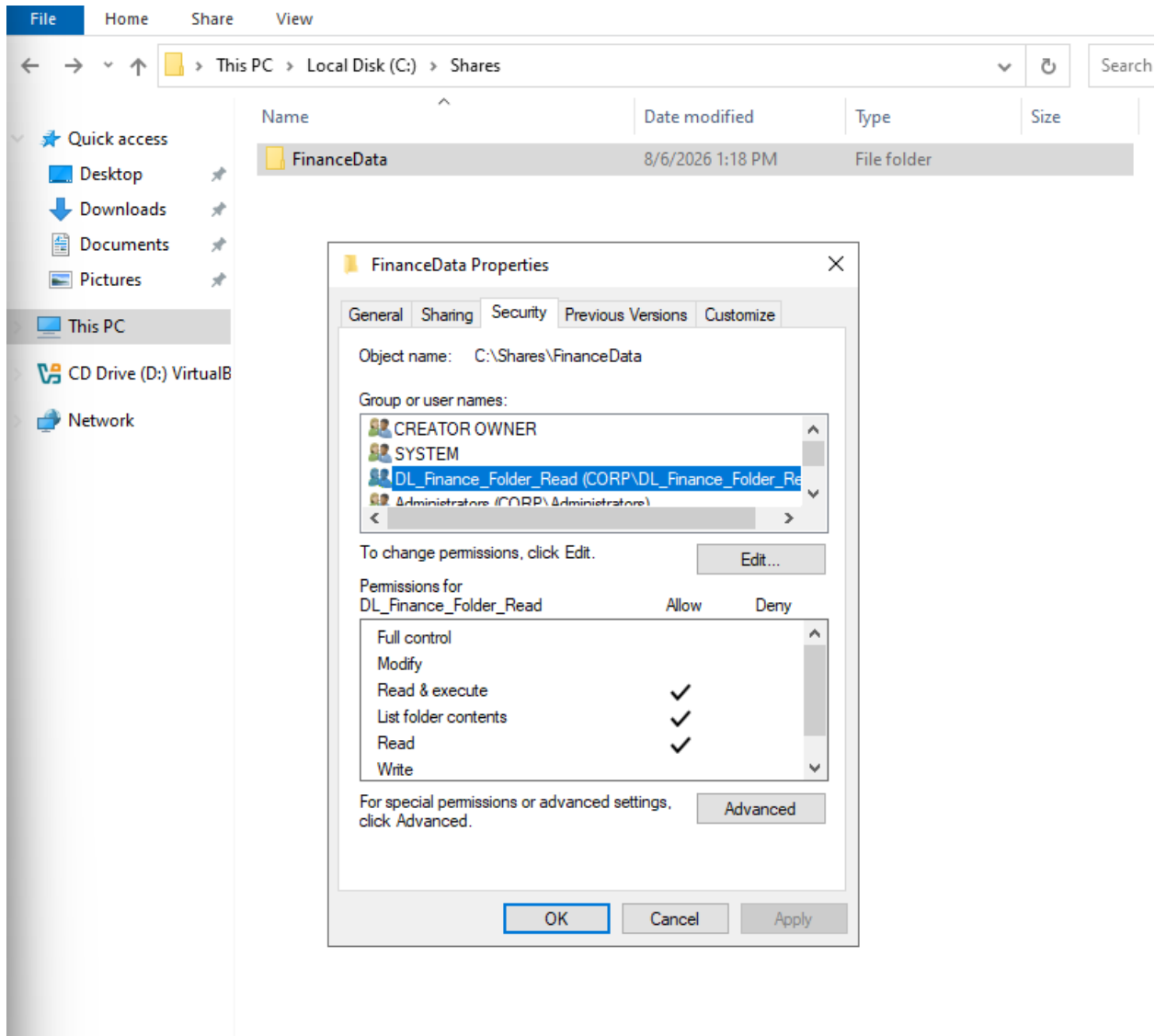
Mode                LastWriteTime         Length Name
----                -
d-----          8/6/2026   1:11 PM                IAM_Scripts
Successfully Provisioned: jdoe in OU=Finance,OU=Corporate_Users,DC=CORP,DC=LOCAL
Assigned jdoe to Security Group GG_Finance_Read
Successfully Provisioned: jsmith in OU=HR,OU=Corporate_Users,DC=CORP,DC=LOCAL
Successfully Provisioned: arivera in OU=Engineering,OU=Corporate_Users,DC=CORP,DC=LOCAL

PS C:\Users\Administrator> _
```

4. Role-Based Access Control (RBAC) & Folder ACL Enforcement

- Applied Least Privilege access control on the C:\Shares\FinanceData enterprise SMB share by granting NTFS Read & Execute permissions exclusively to the DL_Finance_Folder_Read Domain Local group.

```
Administrator: Windows PowerShell
PS C:\Users\Administrator> # 1. Create Share Directory
> New-Item -Path "C:\Shares\FinanceData" -ItemType Directory -Force
>
> # 2. Expose SMB Share
> New-SmbShare -Name "FinanceData" -Path "C:\Shares\FinanceData" -FullAccess "Everyone"
>
> # 3. Enforce NTFS Permissions for Domain Local Group
> $Acl = Get-Acl "C:\Shares\FinanceData"
> $Ar = New-Object System.Security.AccessControl.FileSystemAccessRule("CORP\DL_Finance_Folder_Read", "ReadAndExecute", "ContainerInherit, ObjectInherit", "None", "Allow")
> $Acl.SetAccessRule($Ar)
> Set-Acl "C:\Shares\FinanceData" $Acl
```



5. Automated User Deprovisioning & Access Revocation

- Executed an automated offboarding script to enforce Zero Trust identity management by instantly revoking group memberships (GG_Finance_Read), disabling the account (jdoe), and moving the identity object to the Disabled_Users Organizational Unit.

```
Administrator: Windows PowerShell
PS C:\Users\Administrator> # Offboard John Doe (jdoe)
PS C:\Users\Administrator> $User = "jdoe"
PS C:\Users\Administrator> Disable-ADAccount -Identity $User
PS C:\Users\Administrator> Remove-ADPrincipalGroupMembership -Identity $User -MemberOf "GG_Finance_Read" -Confirm:$false
PS C:\Users\Administrator> Move-ADObject -Identity (Get-ADUser $User).DistinguishedName -TargetPath "OU=Disabled_Users,DC=CORP,DC=LOCAL"
PS C:\Users\Administrator> Write-Host "Successfully Offboarded: $User (Account Disabled, Groups Revoked, Moved to Disabled_Users OU)" -ForegroundColor Red
Successfully Offboarded: jdoe (Account Disabled, Groups Revoked, Moved to Disabled_Users OU)
PS C:\Users\Administrator>
```

Powershell Automation Scripts	
User Onboard Engine	<pre> Import-Module ActiveDirectory \$Users = Import-Csv -Path "C:\IAM_Scripts\new_employees.csv" foreach (\$User in \$Users) { \$SAM = (\$User.FirstName.Substring(0,1) + \$User.LastName).ToLower() \$OU = "OU= \$(\$User.Department),OU=Corporate_Users,DC=CORP,DC=LOCAL" # Production note: Dynamically generate passwords or pull from PAM/Vault APIs \$SecurePassword = ConvertTo-SecureString "TempP@ssword2026!" -AsPlainText - Force if (-not (Get-ADUser -Filter "SamAccountName -eq '\$SAM'")) { New-ADUser -Name "\$(\$User.FirstName) \$(\$User.LastName)" `             -GivenName \$User.FirstName ` -Surname \$User.LastName `             -SamAccountName \$SAM ` -UserPrincipalName "\$SAM@CORP.LOCAL" `             -Path \$OU ` -AccountPassword \$SecurePassword `             -Enabled \$true ` -ChangePasswordAtLogon \$true `             -Title \$User.Title ` -Department \$User.Department Write-Host "Successfully Provisioned: \$SAM in \$OU" -ForegroundColor Green if (\$User.Department -eq "Finance") { Add-ADGroupMember -Identity "GG_Finance_Read" -Members \$SAM Write-Host "Assigned \$SAM to Security Group GG_Finance_Read" - ForegroundColor Yellow } } } </pre>
User Offboarding Routine	<pre> \$User = "jdoe" Disable-ADAccount -Identity \$User Remove-ADPrincipalGroupMembership -Identity \$User -MemberOf "GG_Finance_Read" -Confirm:\$false Move-ADObject -Identity (Get-ADUser \$User).DistinguishedName -TargetPath "OU=Disabled_Users,DC=CORP,DC=LOCAL" Write-Host "Successfully Offboarded: \$User (Account Disabled, Groups Revoked, Moved to Disabled_Users OU)" -ForegroundColor Red </pre>

Lessons Learned

- AGDLP Architectural Decoupling: Implementing the AGDLP strategy demonstrated how separating user identities from resource permissions prevents access control clutter. Nesting GG_Finance_Read into DL_Finance_Folder_Read ensures scalability without manually editing NTFS Access Control Lists whenever users join or leave.
- Automated Identity Lifecycles: Manual user management introduces administrative lag and configuration drift. Automating provisioning and offboarding via PowerShell enforces strict naming conventions (sAMAccountName), automates initial password hygiene, and eliminates orphan accounts during employee departures.
- Zero Trust & Least Privilege: Enforcing strict permissions on enterprise SMB shares (C:\Shares\FinanceData) limits lateral movement and guarantees that users only maintain access strictly required for their job function.
- Rapid Incident Containment: Automated offboarding scripts demonstrate how access can be instantly severed during emergency terminations by disabling the account, revoking active group memberships, and shifting the identity object to a restricted Disabled_Users OU in a single execution.

Conclusion

This project successfully establishes an automated, scalable, and secure Identity and Access Management framework within an Active Directory environment (CORP.LOCAL). By pairing standardized Organizational Unit hierarchies with AGDLP role-based access control and PowerShell lifecycle automation, the infrastructure minimizes security risks associated with privilege creep and unauthorized file access. The implementation demonstrates end-to-end identity management principles essential for enterprise security, regulatory compliance, and efficient IT operations.